

1 Connection of the HYDRA Time and Attendance to SAP R/3 HR

1.1 Interface technology

HYDRA directly communicates with the SAP system using the "RFC technology" and ensures the correct exchange of IDocs.

The document "MYSAP communication configuration" describes how to set up the RFC connection to the SAP server. The document describes the interfaces relevant to HR-PDC in section "Configuration of the distribution model".

1.1.1 Types of data provision

In HR-PDC, SAP always **provides basic data** to the subsystems. Delta data and deleted data are not provided.

When the basic data is provided, the old data is retained and internally marked as "old". Then the different data records are read and each marked as "current". To complete the provision of basic data, the remaining "old" data records are definitively locked or deleted. This ensures that with each provision of basic data, a valid data set is available in the HYDRA system at any time.

1.2 Processing of information fields

The interface for time balances (HR-PDC from SAP R/3 Release 4.5A) includes variable information fields. There are two ways to process the info fields: "Interpretation as account balances" (flextime, leave and other balances) and "Simple transfer of the field values" for display on the terminal.

MPDV can use an environment variable to control which of the options is activated. When the system is implemented, the more comprehensible of the two options, "Interpretation as account balances," is activated. With this option, a maximum of 8 out of 10 information fields passed can be displayed. The names of the information fields can be set in the [Configuration of Accounts](#). If you require the second option, because data that cannot be interpreted as account balances must be displayed, then please contact our hotline.

1.2.1 Interpretation as account balances

HYDRA interprets the data entered in the info fields as account balances and transfers this data to the current HYDRA accounts. These can then be displayed on the PZE terminal using the INFO function.

Assignment (the standard designations of the HYDRA time accounts are given in brackets):

INFO1	Account 1 (flextime)
INFO2	Account 2 (overtime)
INFO3	Account 3 (flextime)
INFO4	Account 4 (leave)
INFO5	Account 5
INFO6	Account 6
INFO7	Account 7
INFO8	Account 8
INFO9	Not processed.
INFOA	Not processed.

You can configure the layout of the information displayed on the terminal on the MOC in the [Configuration of Accounts](#):

- You can configure any account designation.
- You can change the sorting of the accounts.
- You can control the accounts that are displayed on the terminal.
- Formatting can be done in hours or days.



If the configuration of the accounts has been changed, it can happen that the previous account contents are incorrectly interpreted and displayed. The display will be corrected, once the time balances have been downloaded again from SAP.

On the terminal, the designation is displayed in front of the account value, which is helpful for the terminal users. This clear display also promotes acceptance of the time and attendance system. The account values are also available in the application [Current account balances](#) and can be displayed there.

Valid number formats in the interface file

Format	For time accounts	For day accounts
xxx:xx Separator: colon	Time in standard minutes (divided by 60)	-
xxx.xx separator: point or xxx,xx separator: comma In both formats, the number of decimal places is unlimited. One or two decimal places are useful.	Time in industrial minutes (divided by 100)	Days. The number of decimal places saved in HYDRA depends on the factor selected in the configuration of the HYDRA accounts.

If a minus sign ("-") appears in the info field, the value is interpreted as negative. It does not matter on what side the minus sign stands.

Examples of valid formats

" 123.12-"	" 123:12-"
"- 123.12"	" 123.12"
" -123.12"	" 123.1"
"123.12 -"	"123.1 "

The evaluation status can be entered as date in one of the information fields (also INFO9 or INFOA). HYDRA automatically identifies this status. The status is displayed in the HR master data. Activate the display of the evaluation status on the PZE terminal in the [Terminal configuration](#).

Valid formats for the evaluation status in an information field are:

- DD.MM.YY
- DD.MM.YYYY

1.2.2 Simple transfer to the terminal

When transferred, the contents of the 10 info fields are displayed on the terminal without any interpretation or caption. An advantage of this method is that field values that cannot be interpreted as balance are also displayed on the terminal. This could be a text, for example. Another advantage is that all 10 information fields can be displayed.

1.3 Mail IDs IMAIL and ZMAIL

If you use the two mail IDs, "Mail ID time evaluation error" (IMAIL) and "Mail ID time recording type" (ZMAIL) and you want to display an info text, then you must configure the text in HYDRA PZE for each mail ID and each BDE group (any text is possible). This configuration is described in the section that follows.

1.4 Configurations in HYDRA

Not all settings and configurations required in HYDRA are included in the interface and can be taken over from SAP. The following configurations have to be performed in HYDRA. However, the settings can only be made for master data that does not frequently change.

1.4.1 Terminal configuration

Terminals have to be configured based on the current documentation of [Terminals](#) and HYDRA-PZE.

Note: If a terminal is configured with the operation mode "auto status", auto status clocking records are transferred to SAP using the record type "P01 auto status (clock-in or clock-out)" OR using record type "P10 clock-in" or "P20 clock out"; which record type is used depends on the customization options of the interface provided for the upload of time events.

1.4.2 Terminal groups

The SAP HR master includes an access control group. This group must exist as HYDRA [Terminal group](#). You can then create and assign any clocking authorizations groups in SAP. As minimum configuration, all existing terminals have to be assigned to the terminal group "99". Please also see the description of the field `SUBSYSTEM_GROUPING` in the description of the HR master interface in one of the sections below.

1.4.3 Authorization for responsibility areas

To view and edit staff data in HYDRA, you must assign authorizations for responsibility areas to the relevant [Persons](#) on the HYDRA client.

1.4.4 Message configuration for IMAIL and ZMAIL

The HR-PDC interface transfers the ZMAIL field when downloading the HR master. The mail ID of the time recording "IMAIL" is included when time balances are downloaded.

1.4.4.1 Editing in SAP

- **Editing in info type 50 of the SAP HR master data:**

You can store a mail ID for each person in the info type 50 of the SAP HR master data. This ID is transferred when the HR mini-master is downloaded from SAP to HYDRA.

For details on the setting options, see the [SAP help](#). →

- **Integration into the time calculation (balances)**

If errors occur during time calculation, a mail ID can be assigned to the error. To make the required configurations as customizations, go to *SPRO → Personnel Time Management → Time evaluation → Time evaluation based on time → Output of messages → Create message description*

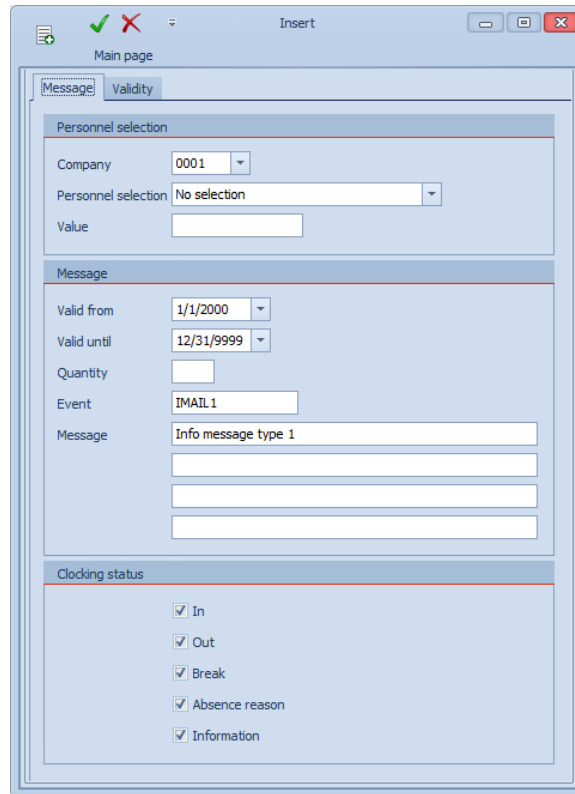
1.4.4.2 Editing in HYDRA



The option "PZE as SAP subsystem" must be activated in the HYDRA basic settings. Only then, the message configurations for "events" starting with "ZMAIL" or "IMAIL" are correctly processed. If the option "PZE as SAP subsystem" is not activated, these configurations are processed as normal messages and a message is created on the terminal for all persons of the

company.

If you use the "mail ID time evaluation error" (IMAIL, from download of balances) and the "mail ID time recording type" (ZMAIL, from SAP HR master), you require a configured message in HYDRA-PZE for each mail ID and each company used. Only then, a configured info text can be displayed:

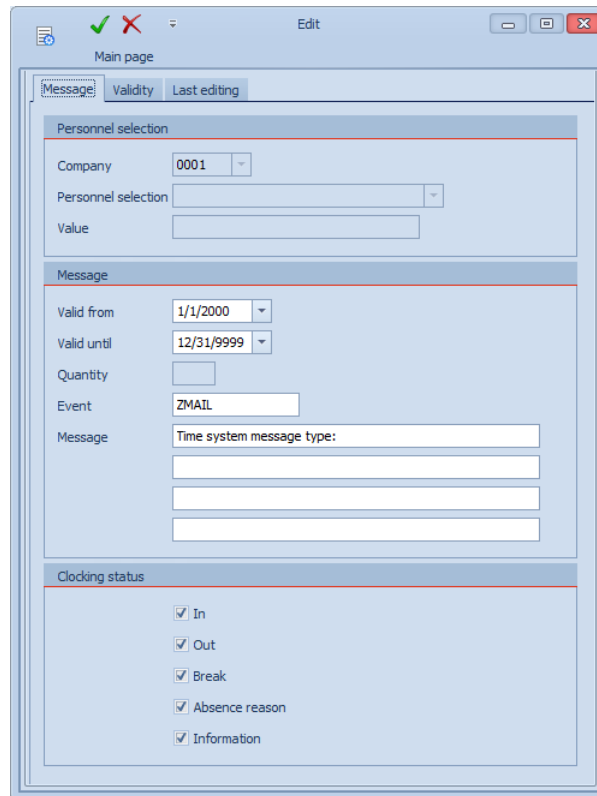


The screenshot shows a software window titled "Main page" with a toolbar at the top containing a green checkmark, a red X, and an "Insert" button. The window is divided into three main sections:

- Personnel selection:** Contains a "Company" dropdown menu set to "0001", a "Personnel selection" dropdown menu set to "No selection", and a "Value" text input field.
- Message:** Contains a "Valid from" date dropdown set to "1/1/2000", a "Valid until" date dropdown set to "12/31/9999", a "Quantity" text input field, an "Event" dropdown menu set to "IMAIL 1", and a "Message" text input field containing "Info message type 1". Below this are three empty text input fields.
- Clocking status:** A list of checkboxes, all of which are checked: "In", "Out", "Break", "Absence reason", and "Information".

Configuration example for the IMAIL ID "1" from download of balances

You can also configure general messages for IMAIL and ZMAIL without attaching a key. If no message is available with the relevant key, this general message will automatically be displayed with the attached key:



Example of a general configuration for the ZMAIL ID from the HR master

If a person is supposed to receive a message with the key "W" and a separate message (event "ZMAILW") is not available for this key, the terminal displays the message "Time mail with ID: W" for the configuration shown in the above screenshot. If no message configuration at all is available, a standard message is generated. Messages of the time recording are displayed with the text "Info mail: X", other messages show the text "personnel mail: X". These standard messages are only displayed on the terminal when the "info" or "messages" keys are used, but not for IN/OUT clockings. Errors are not logged if configurations are missing.



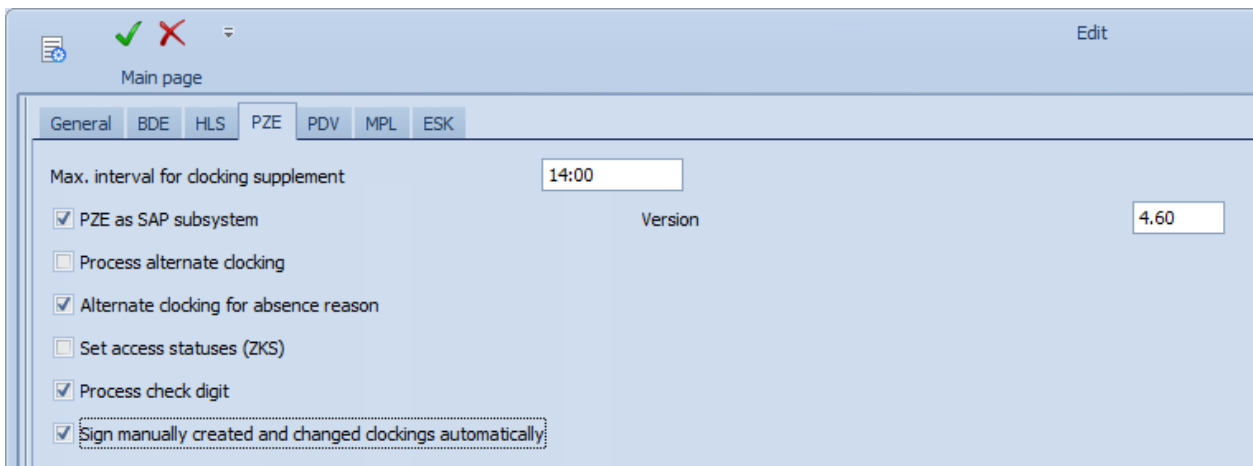
Besides the configuration for IMAIL and ZMAIL, you can also create "normal" messages in HYDRA that are displayed on the terminal. For these messages, the field *Event* must remain empty.

1.4.5 Function authorizations

If you restrict the [Function authorizations](#), you can assign useful access rights in the PZE menu to the employees working on the HYDRA client.

1.4.6 Basic settings

You must activate the option "PZE as SAP subsystem" in the basic settings and enter the relevant version:



For *Version*, enter the version of the SAP system. If you connect via HR-PDC, the version must be 4,50 or higher. Higher version numbers have the same effect.

Most of the changes in the basic settings only become effective after restart of the HYDRA server.

1.4.7 Absence reason "business trip"

If you use HR-PDC and you want to post business trips, you must create an [Absence reason](#) absence reason "business trip" and assign this reason to one of the keys on the [Terminal](#).

The following settings are relevant for the absence reason (all other fields are not relevant):

Absence reason

Should be "1" as this absence reason is specified by default if specific third-party terminals are used.

Designation

"Business trip"

Meaning

Business trip



To correctly process business trips, the option "PZE as SAP subsystem" with version 4.5 or higher must be set in the PZE tab of the *Basic settings*. If these settings are not correct, business trip postings are only transferred as clock-out or clock-in to SAP.

1.5 Useful functions of HYDRA Time & Attendance

The following functions may be used if the time and attendance module (PZE) is used as subsystem of SAP:

- Edit HR master (e.g. "department" field).
- [Clocking authorizations](#) for PZE terminals using terminal groups or terminals
- Control of access authorizations for the HYDRA Access Control system via [Access profile assignments](#) (with ZKS-VWF license only).
- Display of clocking records performed on HYDRA terminals using the functions [Clockings](#) and [Clocking archive](#) (with the PZE-EPP license only).
- Current [Overviews of attendance and absence](#) (with the PZE-INF license only).
- Overview of the [current account balances](#) transferred from SAP (with the PZW-PAP license only).
- Display of the account balances transferred from SAP on the terminal.
- Display of [Messages](#) on the terminal (with PZE-INF and specific terminal types only).
- [Print of ID cards](#) with HYDRA (with SIS-DMA license only).

1.6 Overview of the HR-PDC interfaces

The below overview shows the interfaces and the programs and log files that are used for operation. You can view and print the log files on the MOC in "System administration - Logging - System logs". They should be checked occasionally.

Program	Function	Log files
sap45ein.out	Downloads from SAP to HYDRA: HRCC1DNPERSO: HR master HRCC1DNBALAN: Time balances HRCC1DNTEVGR: Time event groups HRCC1DNATTAB: Absence reasons HRCC1DNOBJID: Objects HRCC1DNCOSTC: Cost centers HRCC1DNDNINRD: Internal orders HRCC1DNWBSEL: Projects HRCC1DNEXTWT: External wage types	dnperso dnbalan dntevgr dnattab dnobjid dncostc dninord dnwbsel dnnextwt
sap45rck.out	Returns uploads from PZE clocking records to HR	sap45rck.pro sap45rck.err

1.6.1 Download HR master

Field (* = key)	name	Pos. / width	Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS		1 10	Logical system, source system	Saved in info text 16.
* TIMEID_NO		11 8	Badge number	Badge number (must be unique)
* FROM_DATE		19 8	Start date	The employee's current HR master data is identified using the validity period. The date of joining in HYDRA is only filled when data is first transferred from SAP and results from the start date. (Format YYYYMMDD).
* TO_DATE		27 8	End date	
* TIMEID_VERSION		35 1	Badge version	Not processed.

PERNO	36	8	Personnel number	Personnel number (must be unique in HYDRA!)
EDIT_NAME	44	40	The person's formatted name (first name, last name, in capital/lower case letters)	Not processed by default
SORT_NAME	84	30	Sortable name (second name, first name, everything in upper case, no umlauts)	Name of employee, is saved as second name. If the name contains a comma, the program assumes that the second name comes before and the first name after the comma, and saves the name with separated first and second names.
LANGU	114	1	Language key	Not processed. In HYDRA, only the ISO language key is integrated.
LANGU_ISO	115	2	Language key according to ISO code 639	Language key for the allocation of the correct texts. Saved as nationality.
PS_GRP_ATT_ABS_TYPE	117	2	Personnel groups attendance/absence	Stored in text field 19 [1,2] and employee subgroup [4,5]
COUNTRY_GROUPING	119	2	Country groups	Stored in text field 17.
SUBSYSTEM_GROUPING	121	3	Groups for sub-system connection	Terminal group for PZE access authorization. Also saved in the "Department" field of HR master data.
ES_GRP_WORK_SCHED	124	1	Groups for working time plan	Stored in text field 18 and employee subgroup [6].
ACCESS_CONTROL_GROUP	125	2	Access control group	Access profile for HYDRA access control system.
PERSONAL_CODE	127	4	Personal code	PIN code of ZKS badge. By default, this PIN code is included in the badge, but not in the HR master. See also the customization options below.

MAIL_INDICATOR	131	1	Mail ID time recording info type	Generation of a relevant message on the terminal.
ATT_ABS_REASON_GRP	132	3	Groups of attendance/absence reasons	Stored in text field 19 and in characters 1 to 3 of the "Employee sub-group" field in the HR master data.
EXT_WAGETYPE_GRP	135	3	Groups of external wage types	Stored in text field 20.
TIME_EVENT_TYPE_GROUP	138	2	Groups of time event types	Time event authorization, stored internally (invisible field "PZT number of").
COMP_CODE	140	4	Company code	Company
COSTCENTER	144	10	Cost center	Cost center. Leading zeros are cut off with numeric cost centers, to save typing work in HYDRA and reduce the number of characters.
CUSTOMER_FIELD_1	154	20	Customer-specific field 1	Reserved for customer-specific processing, stored in info text 14.
CUSTOMER_FIELD_2	174	40	Customer-specific field 2	Reserved for customer-specific processing, stored in info text 9.

Employees who are available in HYDRA, but are no longer contained in the current interface file, are blocked in HYDRA and not deleted.



The badge number and the personnel number must both be unique in HYDRA. This also applies to the badge numbers of blocked employees, which can only be assigned once in HYDRA. If duplicate badge numbers occur, this is recorded in the error log.

For further information on the HR master fields, refer to the documents of the standard interface of the HYDRA Personnel Time Management and the documents of the HR master on the HYDRA client.

1.6.1.1 Access Control Group – Clocking Authorization

The field “groups for subsystem connection” (SUBSYSTEM_GROUPING) is interpreted in HYDRA as a terminal group for the **clocking authorizations at the time and attendance terminals**. The employee is assigned a [clocking authorization](#) for the transferred terminal group. If no valid value is entered, the terminal group 99 is assigned to the person. The assignment of terminals to [terminal groups](#) must be edited in HYDRA.

The access control group (ACCESS_CONTROL_GROUP) is interpreted as an **access profile for the HYDRA access control system**. If no valid value is entered, the access profile 999 is assigned to the employee’s badge. The [Access profiles](#) and their [Access authorizations](#) must be edited in HYDRA.

According to our information, the access control group can be edited in the SAP info type 50 of the HR master.

1.6.1.2 What data is updated?

In some HYDRA systems, you must manually edit the data of persons downloaded from SAP. You must assign a BDE authorization, for example. You can also manually create persons that add to the persons downloaded from SAP. Their personnel and badge numbers must not be identical to numbers of the persons managed in SAP.

With this type of systems, it is important to know the data that is updated during an SAP download (i.e. overwritten) and how the system reacts in case of double personnel or badge numbers. The following rules apply:

HR master data

In a standard system, the following fields of an existing HR master are overwritten by the SAP HR master data interface:

- Badge
- Company
- Area
- Employee subgroup
- Cost center
- Last name, first name
- Business trip authorization (a fixed value is predefined, it is enabled by default. See also the customization options)
- Blocking ID
- Date of leaving
- Nationality

- Free info text 9 (customer field 2)
- Free info text 14 (customer field 1)
- Free info text 16 (source system)
- Free info text 17 (country grouping)
- Free info text 18 (ES_GRP_G_WORK_SCHED)
- Free info text 19 (PS_GRP_G_ATT_ABS_TYPE + ATT_ABS_REASON_GRP_G)
- Free info text 20 (EXT_WAGETYPE_GRP_G)
- Responsibility area (only if activated via additional customization)

When a person is first transferred to HYDRA, the start date from the interface is assigned to the date of joining field in addition to the fields mentioned above.

If an employee is no longer transferred from SAP to HYDRA, then the HR master data is blocked in HYDRA (not deleted). If the person is then downloaded again from SAP, the HR master data is released and the employee's data kept in HYDRA are retained. If the version control of the HR master is used, only versions are blocked that are valid from now on in future. Older HR master versions even remain active in HYDRA if they are no longer included in the interface, as SAP does not provide a complete history.

The date of leaving remains empty in all HR master versions by SAP.



If the interface contains an employee whose badge number is already allocated to a different personnel number in the HYDRA HR master data, the other person's badge number is removed (empty field). A new version is created and the badge number is kept for the past validity period of the person that overlaps with the person included in the interface. For information on the ZKS badge, refer to the next section "ZKS badge".

HYDRA supports HR master versions. The different versions of the HR master in the interface are therefore correctly integrated in HYDRA. A new version is only formed in HYDRA when the data transferred by the interface is different to the previous version.

PZE access authorizations

The access authorization transferred by SAP is always created. PZE access authorizations that are managed by the SAP interface can be identified by the field "user". With access authorizations from the SAP interface, this field includes the name of the logical SAP system that provided the authorization.

Additional PZE access authorizations can be created manually and are only changed by the interface if they are valid for the same terminal group as specified by the interface.

ZKS Badge

In the ZKS badge, the following fields are overwritten by the HR master data download from SAP:

- Name
- Company
- Personnel number
- Validity date
- Comment 1
- Comment 2

Only the data in employee badges are updated.

Other badges can be manually maintained for the employee as replacement badges, but these are not affected by the HR master data download from SAP.

If an employee's badge number has been changed, the old badge is deactivated (blocked) and a new one created, and the existing profile allocation is transferred to the new badge and deleted for the old badge.

If there is an active ZKS badge with this badge number, but a different personnel number, then the badge will be assigned to the new personnel number. The profile assignments of the badge are deleted as they did not belong to the concerned person.

If an employee is no longer transferred from SAP to HYDRA, then the ZKS badge is blocked (not deleted) in HYDRA. If the employee is then downloaded again from SAP, a new ZKS badge is created which retains the previous allocation to the access profile.



The manually maintained data of a normal ZKS badge is lost, if an employee was not included in a download (therefore blocked) and is then transferred again in a later download. In this case a new ZKS badge is created. If the badge number is the same, however, the access profile allocation is still retained.

HYDRA creates new versions of the ZKS badges in the interface for every version of the HR master data that includes differences. With each interface run, it is ensured that the currently valid badge is activated.

ZKS Profile Assignments

With profile assignments, HYDRA memorizes in an internal database field of profile assignments which profile assignments were made by the interface. These assignments are checked and edited with each interface run. The internal database field is also retained when profile assignments are manually changed and copied. The badges are then still managed by the interface.

Additional ZKS profile assignments can be created manually and are not affected by the interface.

1.6.1.3 Configuration options

The following configuration options are available for the HR master download using the interface program sap45ein.scr. Set these options in the entries of the [HYDRA Distribution model](#).

/VAB=x : Responsibility area from field x (x stands for the below values)

This option defines how the "responsibility area" field is assigned in the HR master:

x=KST-FIR : combination of cost center and company, downward compatible with previous cost center authorizations.

Otherwise : empty at first, can be edited manually.

/ZKSPROF=x : ZKS access profile from field x (x stands for the below values) (previously -Hx)

/PZEZUG=x : PZE access authorization from the field x (x stands for the below values) (previously -Tx)

Fields for clocking authorization (PZE) and access authorization (ZKS):

x=B : SUBSYSTEM_GROUPING (BDE group) (standard for PZE)

x=Z : ACCESS_CONTROL_GROUP (access control group) (standard for ZKS)

x=P : PERSONAL_CODE (personal code)

x=T : ES_GRPG_WORK_SCHED (grouping of working time plan)

x=Cn: CUSTOMER_FIELD_n (customer-specific field n, n =1,2)

/DESTCOMP=x : In HYDRA all persons are assigned to the target company x . (previously -Fx)

Using this option, you can set the same entry in field *Company* for all persons in the HYDRA HR master, that are transferred by this interface. In this case, the COMP_CODE included in the HR master interface from SAP is not transferred as the company. If this option is set, the interface does not edit persons that are not assigned to the target company.

If **/DESTCOMP=SSGR** is set, the HYDRA company is transferred in a way that is compatible with KK1 from the BDE group (SUBSYSTEM_GROUPING), instead of the COMP_CODE.

/MAILINDIC : Process mail ID in the HR master and info (previously -M)

If this ID is set, the SAP mail IDs included in the SAP HR master and the balances will be processed. If the SAP mail ID is set, the PZE terminal will show an information when clocking.

/NOBER : Do not overwrite HR master field "area". (previously -NB)

This option makes sure that the field "area" will not be overwritten by the HR master interface for persons that already exist in HYDRA. But it may be edited manually. The field is assigned to "0" if the HR master is transferred for the first time.

/DGBER=J/N : Disable business trip authorization in the HR master.

This option specifies whether or not the authorization for the business trip in the HR master is set by the interface. By default, the business trip authorization is enabled.

/LOGSYS=X : Only process data of the SAP source system X

(with several independent source systems). If data derives from other source systems, it is ignored.

/CUS1=X : HR master customer-specific Field 1 (C20) (values for X see below)

X=TAETIGKEIT : Transferred to the field "activity".

or

X="" : Transfer to field "INFOFELD_14".

/CUS2=X : HR master customer-specific Field 2 (C40) (values for X see below)

X=TAETIGKEIT : Transferred to the field "activity".

or

X="" : Transfer to field "INFOFELD_9".

/NOPCODE : Do not overwrite PIN code in the ZKS badge.

The PIN code from the HR mini master is not transferred to the ZKS badge. By default, the PIN code of the badge will always be overwritten by the PIN code from the interface. The PIN code in the badge controls check of the PIN code in the ZKS access control.

/PIN_PNR : Transfer PIN code to the HR master

The PIN code from the HR mini master is transferred to the HR master. By default, the PIN code of the HR master is NOT overwritten by the interface. The HR master PIN code controls login authorizations within HYDRA@Web and for postings by persons with mobile data acquisition devices.

/ENAME : Take over names from the field EDIT_NAME instead of SORT_NAME.

The person's name is taken from the field EDIT_NAME instead of the SORT_NAME field. The EDIT_NAME field includes the name written with capital/lower case letters and umlauts. The first name is entered before the last name. Please note: The SAP interface does not clearly distinguish between first name and last name. For this reason, HYDRA splits up the name into first name and last name as of the first blank character. However, this might result in several first names to be presented in an uncommon manner, if these names do not include a hyphen in SAP-HR (correct: "Anne-Catherine Smith").

/ENAME_NOSEP : Transfer names from the field EDIT_NAME instead of the SORT_NAME field to the last name.

Corresponds to the /ENAME option. But HYDRA does not distinguish between first name and last name and stores the entire name in the last name field. As a consequence, you can no longer sort by the last name of a person in HYDRA, but only by the first name of a person.

Default value for the gender of a person

A person's gender is not included in the interface. From service pack 16, the default value for the HR master in our system is "not defined". In former versions, the default value used to be "male", because "not defined" was not yet available.

If you want to specify another default value than "not defined", you can configure a deviating default value using an INI configuration as of service pack 16:

INI configuration

PNR/DEFAULTS/GENDER

Content

Contents	Meaning
M	Male
W	Female
X	Third gender
U	Not defined

1.6.2 Download of time balances

The time balances are allocated to the employees and can be viewed on all PZE terminals and on the MOC using the info function.

For further information on the display options for time balances on PZE terminals, refer to section "Processing of the info fields".

Field name (* = key)	Pos.	/ width	Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS	1	10	Logical system, source system	Reference to HR master, info text 16
* TIMEID_NO	11	8	Badge number	Reference to the badge number in the HR master
PERNO	19	8	Personnel number	-
SUBSYSTEM_GROUPING	27	3	Groups for sub-system connection	-
INFOFELD_1	30	13	Variable information field	Value account 1
INFOFELD_2	43	13	Variable information field	Value account 2
INFOFELD_3	56	13	Variable information field	Value account 3

INFOFELD_4	69	13	Variable information field	Value account 4
INFOFELD_5	82	13	Variable information field	Value account 5
INFOFELD_6	95	13	Variable information field	Value account 6
INFOFELD_7	108	13	Variable information field	Value account 7
INFOFELD_8	121	13	Variable information field	Value account 8
INFOFELD_9	134	13	Variable information field	-
INFOFELD_10	147	13	Variable information field	-
TIME_EVAL_MAIL_IN DICATOR	160	1	Mail ID time evaluation error	Creation of an appropriate message on the terminal.
CUSTOMER_FIELD_1	161	20	Customer-specific field 1	Reserved for customer specific processing, info text 15.
CUSTOMER_FIELD_2	181	40	Customer-specific field 2	Reserved for customer specific processing, info text 10.

1.6.2.1 Customization settings

The following customization options are provided if you want to customize the download of balances with the interface program sap45ein.scr. Set these options in the entries of the [HYDRA Distribution model](#).

/MAILINDIC : Process mail ID in the HR master and info (previously -M)

If this ID is set, the SAP mail IDs included in the SAP HR master and the balances will be processed. If the SAP mail ID is set, the PZE terminal will show an information when clocking.

/RAWINFO : Information texts as message in the hyinfo table (text form). (previously -I)

If this parameter is set, the information texts from SAP are directly displayed unchanged as information on the PZE terminal. If this option is not set the first 8 of the 10 SAP information fields are imported to the HYDRA time accounts and displayed as time account on the PZE terminal. By default, they are processed as time accounts (option is not set), as SAP only transfers information values without designation or text.

/LOGSYS=X : Only process data of the SAP source system X

(with several independent source systems). If data derives from other source systems, it is ignored.

1.6.3 Download of time event groups

The time event groups are read in HYDRA, but are not used to control clocking authorizations. In HYDRA, this is controlled via access authorizations and terminal configurations.

Field name (* = key)	Pos. / width		Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS	1	10	Logical system, source system	Is saved.
* TIME_EVENT_TYPE_GROUP	11	2	Group of time events	Is saved.
TEVENTTYPE	13	3	Record type	Is saved. See the description of uploading clocking records.

1.6.4 Download of absence reasons

The absence reasons are downloaded so that the relevant names of the absence postings can be displayed on the MOC. Only if the absence reason list is used on the terminal, it is possible to perform a personalized authorization check for absence reason postings.

Field name (* = key)	Pos. / width		Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS	1	10	Logical system, source system	Is saved.
* ATT_ABS_REASON_GRP	11	3	Groups of attendance/absence reasons	Relating to HR master data/Info text_19[3,5] or employee subgroup [1,3]
* PS_GRP_ATT_ABS_TYPE	14	2	Personnel groups attendance/absence	Relating to HR master data/Info text 19 [1,2] or employee subgroup [4,5]
* ES_GRP_WORK_SCHEDULE	16	1	Groups for working time plan	Relating to HR master data/Info text 18 or employee subgroup [6]
* ATT_ABS_REASON	17	4	Attendance/absence reason	Absence reason
* FROM_DATE	21	8	Start of validity date	Selection of valid data records
* TO_DATE	29	8	End of validity date	Selection of valid data records
* LANGU	37	1	Language key	-
* LANGU_ISO	38	2	Language key according to ISO code 639	Language key for the allocation of the correct texts, relating to HR master data/nationality

ATT_ABS_REASON_TEXT	40	30	Designation of the attendance/absence reason	Designation of the absence reason
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1.6.5 Download of objects

The objects are used for uploads/confirmations via the PZE terminal CT-380 or CT-830. They are only used in very special cases.

Field name (* = key)	Pos. / width		Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS	1	10	Logical system, source system	Is saved.
* OBJECT_TYPE	11	2	Object type	Is saved.
* OBJ_ID	13	8	Object	Is saved.
* OBJ_ID_GRP	21	15	Object grouping	Is saved.
* FROM_DATE	36	8	Start of validity date	Selection of valid data records
* TO_DATE	34	8	End of validity date	Selection of valid data records
* LANGU	42	1	Language key	Is saved.
* LANGU_ISO	43	2	Language key according to ISO code 639	Language key for the allocation of the correct texts, relating to HR master data/nationality
OBJ_ID_TEXT	45	40	Object name	Is saved.

1.6.6 Download of cost centers

The cost centers are downloaded so that the detailed names of the cost centers can be displayed on the MOC.

On the PZE terminal (Windows), the cost centers can also be sent along with other postings, if you assign the function "cost center list" (absence reason KSL) to an absence reason key on the terminal. You can configure a single cost center key with the absence reason KST.

Field name (* = key)	Pos. / width		Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS	1	10	Logical system, source system	Is saved.

* COMP_CODE	11	4	Company code	Company
* COSTCENTER	15	10	Cost center	Cost center
* COSTCENTER_GROUP	25	15	Cost center group	Is saved.
* FROM_DATE	40	8	Start of validity date	Is saved.
* TO_DATE	48	8	End of validity date	Is saved.
COCNTR_TXT	56	20	Cost center designation	Cost center designation

1.6.7 Download of internal orders

The internal orders are used for uploads/confirmations via the PZE terminal CT-380 or CT-830. They are only used in very special cases.

Field name (* = key)	Pos. / width		Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS	1	10	Logical system, source system	Is saved.
* COMP_CODE	11	4	Company code	Is saved.
* ORDER	15	12	Internal order	Is saved.
* ORDER_GRP	27	15	Grouping of internal orders	Is saved.
ORDER_NAME	42	40	Designation of the internal order	Is saved.

1.6.8 Download of projects

The projects are used for uploads/confirmations via the PZE terminal CT-380 or CT-830. They are only used in very special cases.

Field name (* = key)	Pos. / width		Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS	1	10	Logical system, source system	Is saved.
* COMP_CODE	11	4	Company code	Is saved.
* WBS_ELEMENT	15	24	Project ID	Is saved.
*	39	15	Groups of projects	Is saved.

WBS_ELEMENT_GRP				
WBS_SHORTTEXT	54	40	Designation of the project	Is saved.

1.6.9 Download of external wage types

The external wage types are used in very special cases for the upload/confirmation via a special GUI on the MOC. This is not activated by default. If required, please contact our project management department.

Field name (* = key)	Pos. / width		Meaning in SAP	Data field and meaning in HYDRA
* SOURCE_SYS	1	10	Logical system, source system	Is saved.
* EXT_WAGETYPE_GRP	11	3	Groups of external wage types	Is saved.
* COUNTRY_GRP	14	2	Country groups	Is saved.
* EXTERNAL_WAGETYPE	16	4	External wage type	Is saved.
* FROM_DATE	20	8	Start of validity period	Is saved.
* TO_DATE	28	8	End of validity period	Is saved.
WAGETYPE_UNIT	36	3	Unit of external wage type	Is saved.
WAGETYPE_UNIT_ISO	39	3	Unit of the external wage type according to ISO	Is saved.
* LANGU	42	1	Language key	Is saved.
* LANGU_ISO	43	2	Language key according to ISO	Is saved.
WAGELTEXT	45	25	Unit of external wage type	Is saved.
UNIT_TEXT	70	20	Designation of the unit	Is saved.

1.6.10 Upload of time events

1.6.10.1 Record types

The following record types are processed:

Record type	Meaning
P01	Automatic status (IN or OUT) (Note: "automatic status" clocking records are transferred as record type P10 clock-in or P20 clock-out if customized accordingly)
P02	Break automatic status (start or end of break)
P03	Business trip auto status (start or end offsite work)
P04	Work from home auto status (start or end of work from home)
P05	Access log (interim entry)
P10	Clock-in (Note: automatic status clocking records are transferred as record type P01, if customized accordingly)
P11	Change of payment or cost center information
P15	Start of break
P20	Clock-out (Note: automatic status clocking records are transferred as record type P01, if customized accordingly)
P25	End of break
P30	Start of business trip
P35	Start offsite work from home (start offsite work from home)
P40	End of business trip
P45	End offsite work from home
[P50]	[External wage type (employee expenditures), only with UPTEVEN]
P60	Info (info entry)

1.6.10.2 Interface Structure

The time events, which are to be uploaded, result from clocking records made at the PZE terminal.

Field name	Pos. / width		Meaning in SAP	Data field and meaning in HYDRA
SOURCE_SYS	1	10	Logical system, source system	Is saved.
TEVENTTYPE	11	3	Time event type, record type	Record type
TERMINALID	14	4	ID of the shop floor terminal	The terminal number is transferred for events posted on the terminal; the first four letters of the HYDRA user name are transferred for events that are manually entered on the MOC.
LOGDATE	18	8	Date of the event	Clocking date
LOGTIME	26	6	Time of the event	Clocking time
PHYSDATE	32	8	Date when the event was written to the interface	Current date at the time of start of the interface program
PHYSTIME	40	6	Time when the event was written to the interface	Current time at the time of start of the interface program
TIMEID_NO	46	8	Badge number, constant 0.	Badge number, not allocated by HYDRA, the personnel number is used instead.
PERNO	54	8	Personnel number	Personnel number
ATT_ABS_REASON	62	4	Absence reason	Absence reason Note: To transfer absence reasons, you must enter the PZE as HR subsystem in the HYDRA basic settings as of version 4.5.
OBJECT_TYPE	66	2	Object type	SAP object type
OBJECT_ID	68	8	Object ID	SAP object ID
COMP_CODE	76	4	Company key	Company
COSTCENTER	80	10	Cost center	Cost center
ORDER	90	12	Internal order	Internal SAP order (no HYDRA

				operation!)
WBS_ELEMENT	102	24	Project (work breakdown structure)	SAP project
CUSTOMER_FIELD_1	126	20	Customer-specific field 1	Free for customer specific processing
CUSTOMER_FIELD_2	146	40	Customer-specific field 2	Free for customer specific processing

1.6.10.3 Customization settings

The interface program sap45rck.scr prepares the data for the upload of time events. This program is started in cyclic intervals by the HYDRA Scheduler. You can control the behavior/performance, if you set command line parameters in the Scheduler entry.



Options starting with a minus sign must be entered before the options starting with a slash!

-K : Autom. status is not uploaded, clock-in/clock-out are uploaded instead

If this option is not set, HYDRA forwards record type P01 CLOCK_IN_OR_OUT to SAP for auto status clocking records. The record type P03 START_OR_END_OFFSITE_WORK is transferred with business trips. If the option -K is set, HYDRA transfers instead the record types P10 CLOCK_IN and P20 CLOCK_OUT or P30 START_OFFSITE_WORK and P40 END_OFFSITE_WORK.

This option is set by default for newly installed HYDRA systems.

-C <XXX> : Generate upload for a specific SAP client only (SOURCE_SYS) XXX

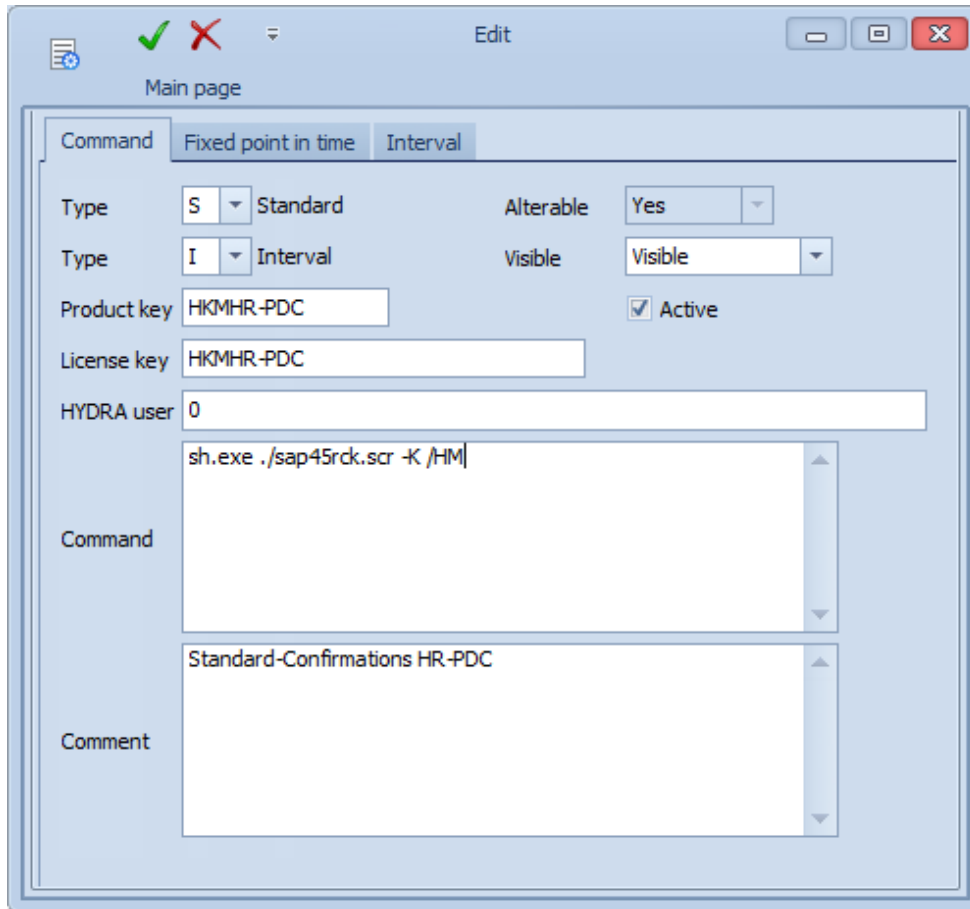
The clocking records of persons transferred from a specific SAP system <XXX> to HYDRA are prepared only. The SAP system is stored in the info field 16 of the HYDRA HR master. If persons from several SAP systems must be processed, you must create several Scheduler entries.

-F <XXX> : Generate upload for a specific company <XXX> only.

The clocking records of persons from company <XXX> are only prepared for being sent to SAP. If persons from several companies are to be transferred to SAP, several Scheduler entries will have to be created.

-B : Generate uploads with segment name E2BPCC1UPT EVEN000.

Time clocking records are uploaded with E2BPCC1UPT EVEN000 instead of the segment name E2BPCC1UPT EVEN.




Options starting with a minus sign must be entered before the options starting with a slash!

/HM : Transfer time of time events exact to the minute without seconds

If the command line parameter /HM is specified, the time of the time events only includes minutes in field LOGTIME. The seconds are cut off and set to "00". The option /HM is available as of HYDRA 8 service pack 10 (2016).

/NOFIRKST: Suppression of cost center and company

The fields for the cost center (COSTCENTER) and company (COMP_CODE) are always left empty and are never filled when uploading clocking records. This option is used, for example, if cost center changes are clocked in HYDRA that are not to be transferred as such to SAP-HR.

1.6.11 Logs and Security Mechanisms

1.6.11.1 Log files

A separate log file is created for each interface. The files are listed in the above tables with the interface overview. You can view and print these files on the HYDRA client in menu item *System administration* → *Logging* → *System logs* (transaction code: syspro).

1.6.11.2 Security mechanisms

The data records created by the upload program are additionally buffered in HYDRA.

In the current versions of the upload/confirmation program, clocking records are stored for around 3 months in a buffer table in the database and can, if necessary, be transferred again. If required, please contact our hotline.

1.7 Connection of Several SAP Systems to HYDRA

Persons can be separated using source system and target company.

If HR-PDC is used, the HYDRA system knows the SAP system (Source-SYS) a person belongs to. This information is stored in the additional data of the HR master data in info field 16:

This ensures, without taking any extra time or effort, that an employee's recorded clockings are confirmed/uploaded to the correct SAP system.

If you want to process HR-PDC persons in the complete HYDRA Time and Attendance, you can perform a controlled processing of specific SAP systems using the Source-Sys or the so called "target companies". The required configurations must deeply affect the HYDRA system and must be performed by MPDV personnel, if required.